Zikang Xu

Date of Birth	Jan., 1997
Expected Graduation Date	July, 2025

Education

School of BME, **University of Science and Technology of China** — Suzhou, China
Ph.D. candidate — Sept.2021 till now
Supervisor：Prof. S. Kevin Zhou (Fellow of IEEE, AIMBE, NAI)

School of BME, **Southeast University** — Nanjing, China
M. Eng. — Sept. 2018 - July 2021
Supervisor：Prof. Zhongze Gu (Fellow of AIMBE, RSC)
Thesis: Microscopic Image Reconstruction and Enhancement Algorithm Based on Deep Learning

School of BME, **Southeast University** — Nanjing, China
Chien-Shiung Wu College (honor college), **Southeast University** — Nanjing, China
B. Eng. — Sept. 2014 - July 2018
Thesis: Machine Learning Based Emotion Recognition of EEG Signals

Research Interests

During my Ph.D. studies, I specialized in **Algorithm Fairness in Medical Image Analysis**, and **Landmark Detection** with my research published (as first and second co-author) in **1** top-tier journal and **3** conferences, including NPJ Dig. Med., MICCAI, and MIDL. I am also interested in **Medical Segment Anything Models**, **Surgical Instrument Segmentation**, and **Biomedical Image Analysis**.

Publications

Algorithm Fairness in Medical Image Analysis

[MICCAI-2023] FairAdaBN: Mitigating unfairness with adaptive batch normalization and its application to dermatological disease classification
Zikang Xu, Shang Zhao, Quan Quan, Qingsong Yao, S. Kevin Zhou

[NPJ Dig. Med.] Addressing Fairness Issues in Deep Learning-Based Medical Image Analysis: A Systematic Review
Zikang Xu, Jun Li, Qingsong Yao, Han Li, Mingyue Zhao, S. Kevin Zhou

[NeurIPS 2024] FairMedFM: Fairness Benchmarking for Medical Imaging Foundation Models
Ruinan Jin*, **Zikang Xu***, Yuan Zhong*, Qingsong Yao, Xiaoxiao Li, S. Kevin Zhou, Qi Dou

[In Preparation] APPLE: Adversarial Protected-attribute-aware Perturbations on Latent Embedding
Zikang Xu, Fenghe Tang, Quan Quan, Qingsong Yao, S. Kevin Zhou

zikangxu@mail.ustc.edu.cn 130-2258-7096 Xuzikang.github.io

[ArXiv] Inspecting Model Fairness in Ultrasound Segmentation Tasks
Zikang Xu, Fenghe Tang, Quan Quan, Jianrui Ding, Chunping Ning, S. Kevin Zhou

Catheter Landmark Detection

[MICCAI-2024] SIX-Net: Spatial-context Information miX-up for Electrode Landmark Detection
Xinyi Wang, **Zikang Xu**, Qingsong Yao, Heqin Zhu, Yiyong Sun, S. Kevin Zhou

[MIDL-2024] OFELIA: Optical Flow-based Electrode LocalIzAtion
Xinyi Wang, **Zikang Xu**, Qingsong Yao, Yiyong Sun, S. Kevin Zhou

Others

[MIDL-2024, Oral] Slide-SAM: Medical SAM Meets Sliding Window
Quan Quan, Fenghe Tang, **Zikang Xu**, Heqin Zhu, S. Kevin Zhou

[MedIA, Major Revision] LACOSTE: Exploiting Stereo and Temporal Contexts for Surgical Instrument Segmentation
Qiyuan Wang, Shang Zhao, **Zikang Xu**, S. Kevin Zhou

[Biosensors and Bioelectronics] A storm in a teacup–A biomimetic lung micro-physiological system in conjunction with a deep-learning algorithm to monitor lung pathological and inflammatory reactions
Zaozao Chen, Jie Huang, Jing Zhang, **Zikang Xu**, Qiwei Li, Jun Ouyang, Yuchuan Yan, Shiqi Sun, Huan Ye, Fei Wang, Jianfeng Zhu, Zhangyan Wang, Jie Chao, Yuepu Pu, Zhongze Gu

Research Experience

Fuwai Hospital, **Chinese Academy of Medical Sciences** — Beijing, China
Research intern — June 2023 - Oct. 2023

Journal and Conference Reviewer

IEEE Trans. Med. Imaging, MICCAI 2024, BIBM 2024

Awards

- 2016 National Biomedical Engineering Innovation Design Competition for College Students
- 2021 Suzhou Industrial Park Scholarship
- 2021,2023 Academic First Class Scholarship in USTC
- 2022 Academic Second Class Scholarship in USTC

Talents

- IELTS 7.0 (Listening: 7.5, Reading: 7.0, Writing: 7.0, Speaking 6.0)
- Table-tennis, Badminton, Photography
- Python, MATLAB, PyTorch, LaTeX